that equalizing the content and distribution of market data provided by exchanges to competing consolidators and self-aggregators and proprietary market data would eliminate the "two-tiered" market data structure.[593] Another commenter stated that the proposal's improvements to content and latency could result in greater reliance on consolidated market data.[594] One commenter stated that the proposal would "put consolidators on more equal footing" with proprietary market data feeds.[595]

Another commenter indicated that the geographic diversification of competing consolidators could increase the use of consolidated market data.[596] The commenter stated that its own need for direct proprietary market data feeds would be eliminated if a competing consolidator were located within the same data center as the broker-dealers the commenter uses.[597] The commenter also stated that it is unlikely that broker-dealers and "higher-turnover market participants" would use consolidated market data as a substitute for lowest-latency, self-aggregated direct proprietary market data feeds because latency minimization is critical for their trading activities.[598]

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593 MEMX Letter at 2.

594 See ICI Letter at 4.

595 Capital Group Letter at 4.

596 See NBIM Letter at 5–6.

597 See id. at 4.

598 Id. at 4, 5. This commenter said that to be "consistently competitive," broker-dealers need to self-aggregate and use the fastest connectivity available. According to this commenter, this would require using direct proprietary market data feeds for algorithmic executions. Id. at 3–4.

The Commission believes that the decentralized consolidation model will modernize the national market system so that consolidated market data is disseminated to market participants in an accurate, reliable, prompt, and fair manner.[599] The Commission also believes that the decentralized consolidation model will help to ensure the accuracy and completeness of consolidated market data.

The Commission disagrees with the comments that stated that the decentralized consolidation model would not achieve the goal of disseminating consolidated market data to market participants in a more timely, efficient, and cost-effective manner than the current centralized consolidation model.[600] Further, in response to comments that stated that competition either would not materialize or would not guarantee any benefits to market participants,[601] the Commission believes that the amendments will allow the introduction of competitive forces, and foster a competitive environment, for the dissemination of consolidated market data. Today, there is no competition in the collection, consolidation, and dissemination of SIP data. The exclusive SIPs do not compete with each other because Rule 603(b) currently requires the dissemination of all consolidated information for an individual NMS stock to occur through an exclusive SIP. Therefore, each exclusive SIP represents different tapes. The amendments to Rule 603(b) will provide an opportunity for competition to improve the dissemination of consolidated market data. Market participants have stated frequently that SIP data is slower than

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599 See 15 U.S.C. 78k-1(c)(1)(B).

600 See Healthy Markets Letter I at 2; Kubitz Letter at 1; Data Boiler Letter I at 46–47; Citadel Letter at 5; TD Ameritrade Letter at 2, 12; NYSE Letter II at 22; Nasdaq Letter IV at 2–3, 8; Angel Letter at 18–20; STANY Letter II at 5.

601 See TD Ameritrade Letter at 12; Data Boiler Letter I at 46–47; Nasdaq Letter IV at 3.

certain proprietary market data products distributed by the exchanges[602] and that the SRO operators of the Equity Data Plans—some of whom have an inherent conflict of interest because their proprietary data products compete with the SIP data distributed by the Equity Data Plans—have had little incentive to improve the quality of SIP data.[603] The exclusive SIPs have not kept pace with the needs of certain market participants, while the exchanges have expanded the content and reduced the latency of their proprietary data products in response to market participants' needs.

Some commenters stated that they will consider entering the competing consolidator business.[604] These statements suggest the potential competitive landscape that will develop in the national market system with the decentralized consolidation model. The Commission believes that competitors will be drawn to the significant market for the enhanced data content that will be included in consolidated market data. By fostering a competitive environment for

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602 See BestEx Research Letter at 1; State Street Letter at 2; SIFMA Letter at 5.

603 See SIFMA Letter at 3, 5; BestEx Research Letter at 1, 4 (citing the Proposing Release, 85 FR at 16767). The first commenter stated that the current market data infrastructure provides no incentives for the SRO operators of the SIPs to make such improvements. See SIFMA Letter at 3. The commenter also noted the "inherent conflicts of interest in the existing exclusive SIP model." Id. at 5.

604 See McKay Letter at 2; MIAX Letter at 1; NovaSparks Letter at 1 ("[A] wide variety of trading firms consolidate this data and we believe several vendors will soon become Competing Consolidator."). See also Miami International Holdings Announces That It Is Evaluating Registration as a Competing Consolidator, dated Nov. 18, 2020, available at https://www.miaxoptions.com/press-releases. The Commission notes that Virtu Financial submitted a comment letter on a proposed rule change in which it expressed interest in establishing a competing consolidator. See letter from Douglas A. Cifu, Chief Executive Officer, Virtu Financial, to Vanessa A. Countryman, Secretary, Commission, dated Aug. 28, 2020, available at https://www.sec.gov/comments/sr-nyse-2020-05/srnyse202005-7707480-222891.pdf.

consolidated market data, the Commission is providing the opportunity for competing consolidators to end the exclusive SIP monopoly by competing on the technology and data services they offer. A competitive environment should lead to the use of new, updated technology in a more expedited fashion than occurs today. The Commission believes that competing consolidators will develop different consolidated market data products and services for their subscribers and will compete on the basis of latency, resiliency, services and products offered, and other factors, including price. The Commission agrees with the commenters who stated that competition will lower market data costs, reduce latency, and provide better quality data.[605]

Due to the structure of the decentralized consolidation model, the Commission believes that competing consolidators and self-aggregators will significantly reduce the geographic, aggregation, and transmission latency differentials that exist between SIP data and proprietary data. With respect to geographic latency, competing consolidators will be able to deliver consolidated market data products directly to subscribers because such data will no longer be required to travel several miles to a separate location for consolidation by the exclusive SIPs. By allowing consolidation to occur at the data center where a data end-user is located instead of occurring only at the CTA/CQ SIP in Mahwah, NJ, and the Nasdaq UTP SIP data center in Carteret, NJ, market participants located outside of these data centers should receive consolidated market data at reduced geographic latencies. With respect to aggregation latency,

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605 See Committee on Capital Markets Regulation Letter at 3; Fidelity Letter at 9; BestEx Research Letter at 1; ACTIV Financial Letter at 1; SIFMA Letter at 5, 12; State Street Letter at 3; IntelligentCross Letter at 5; ICI Letter at 4, 10; RBC Letter at 6; DOJ Letter at 5; Wellington Letter at 1; Capital Group Letter at 4.

competition will incentivize competing consolidators to minimize the amount of time it takes to aggregate SRO data into consolidated market data products. Competition will also incentivize competing consolidators to reduce transmission latency because they will not be restricted to the transmission methods mandated by the Equity Data Plans;[606] therefore, they can compete based on the efficiency of their delivery of consolidated market data products. Even if a competing consolidator chooses not to consolidate data at its users' data centers, the Commission believes the users may still benefit from reduced aggregation and transmission latencies because competing consolidators will be incentivized to use the latest aggregation and transmission mechanisms as a means to attract subscribers.

The Commission believes that the competition fostered by the new model will enhance the speed and quality of the collection, consolidation, and dissemination of consolidated market data. For example, competing consolidators could seek to provide faster consolidation times, reduce transmission and connectivity latency, provide greater connectivity bandwidth, and reduce connectivity fees. Several commenters agreed that competition will enhance the national market system.[607]

The Commission recognizes that some market participants that require the lowest possible latency and additional (e.g., order-by-order data) content may continue to use

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606 As described in the Proposing Release, the transmission methods mandated by the Equity Data Plans typically rely on transmission options that are slower than competitive options. See Proposing Release, 85 FR at 16767.

607 See Committee on Capital Markets Regulation Letter at 3; Fidelity Letter at 9; BestEx Research Letter at 1; ACTIV Financial Letter at 1; SIFMA Letter at 5, 12; State Street Letter at 3; IntelligentCross Letter at 5; ICI Letter at 4, 10; RBC Letter at 6; DOJ Letter at 5; Wellington Letter at 1; Capital Group Letter at 4.

proprietary data feeds for certain trading applications. However, for applications that do not require additional content beyond the scope of new core data, the Commission believes that, once operating in a competitive landscape with the requirement that data be made available "in the same manner and using the same methods, including all methods of access and the same format,"[608] latency alone will not be a compelling reason to subscribe to proprietary data.[609] As affirmed by commenters, the Commission believes the model's significant improvements to the latency and content of consolidated market data products will enhance the usefulness of the data provided to users under the national market system.

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608 Rule 603(b) of Regulation NMS. See infra Section III.B.9.

609 The Commission recognizes that there will be a small "extra hop" for competing consolidators that could result in a small amount of additional latency as compared to proprietary data because competing consolidators must collect, consolidate, and disseminate consolidated market data products to end users. However, the extra hop will be significantly less than the geographic latency that currently exists with the exclusive SIPs. The extra hop refers to the need to transmit data within a data center, a span of feet, as compared to geographic latency among geographically diverse data centers, a span of miles. More specifically, a competing consolidator that chooses to collect, consolidate, and disseminate market data within the same data center as its end-users will only have to disseminate consolidated market data within the data center, while exclusive SIPs must collect data from geographically dispersed SRO data centers, and consolidate and disseminate consolidated market data to end-users in other data centers. If a competing consolidator does not consolidate data at its users' data centers, its end-users may still benefit from reduced aggregation and transmission latencies due to the competitive aspect of the decentralized consolidated model. Further, the amount of latency that may result from using competing consolidators will depend upon other technical choices and competencies of the competing consolidator (i.e., a competing consolidator may choose to use the most technologically advanced aggregation and transmission technologies and therefore narrow its latency differential with proprietary data; conversely, a decision to use less state-of-the-art technology could widen this latency differential while potentially lowering costs to users). Self-aggregators would not have the extra hop because they will be collecting and consolidating this data for themselves. See infra Section III.D.2(d).

The rules adopted for the decentralized consolidation model have been designed to avoid "gaming and fraud"[610] and to ensure the accuracy and completeness of consolidated market data. Competing consolidators and self-aggregators will be regulated entities, which will help monitoring efforts regarding the accuracy and completeness of consolidated market data. Competing consolidators are required to register with the Commission pursuant to Rule 614 and will be subject to Commission oversight. In addition, self-aggregators, which must be registered entities—i.e., broker-dealers, national securities exchanges, national securities associations, or RIAs— will be subject to Commission oversight.

Under Rule 614, all competing consolidators will be subject to standards with respect to the promptness, accuracy, reliability, and fairness of their consolidated market data products' distribution.[611] Form CC will require competing consolidators to provide operational transparency, and Rule 614(d) will require a competing consolidator to publish monthly performance metrics and other information concerning performance and operations.[612] These requirements should help to ensure that consolidated market data products are provided in a prompt, accurate, and reliable manner by providing transparency to subscribers and potential subscribers into a competing consolidator's performance and operations. Because these provisions require that all competing consolidators disclose the same information, they will allow market participants to evaluate and compare competing consolidators more easily based on

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610 Kubitz Letter at 1.

611 Under Section 11A(b)(3)(B) of the Exchange Act, 15 U.S.C. 78k-1(b)(3)(B), the Commission will be able to grant the registration of a competing consolidator only if the Commission is able to find, among other things, that the competing consolidator is so organized, and has the capacity, to be able to assure the prompt, accurate, and reliable performance of its functions and to operate fairly and efficiently as a SIP.

612 See infra Section III.C.8.

cost, service, and performance. These requirements are designed to establish a system whereby a competing consolidator will have to provide consolidated market data products with competitive latency, but also reliably and accurately, and in a cost-effective manner in order to attract and maintain its subscriber base.

The Commission does not believe that the decentralized consolidation model will be impractical or increase market fragmentation. While the decentralized consolidation model introduces multiple competing consolidators disseminating consolidated market data products, today, market participants utilize data products developed by multiple data vendors, exchanges, and the exclusive SIPs. The decentralized consolidation model does not introduce additional fragmentation in the market data landscape.

In the Proposing Release, the Commission described the significant latencies that exist in the centralized consolidation model and explained specifically how the decentralized consolidation model will address them.[613] Further, the Commission discussed how the proposed rules will address data quality, availability, and reliability[614] and the disclosure of fees set by competing consolidators. The Commission provided information demonstrating how the proposed rules would achieve the Commission's goal of modernizing the national market system so that consolidated market data is disseminated to market participants in an accurate, reliable, prompt, and fair manner.

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613 See Proposing Release, 85 FR at 16768. See also supra note 533 (discussing the latencies that exist in the current centralized consolidated model). See also infra Section III.B.5 (discussing comments on the decentralized consolidation model's impact on latency).

614 See Proposing Release, 85 FR at 16782. See infra Section III.C.8 (discussing competing consolidator responsibilities under Rule 614).

3. Comments on the Viability of the Decentralized Consolidation Model

Commenters questioned whether enough competing consolidators would enter the market to make the decentralized consolidation model viable.[615] Some commenters stated that the success of the model depends on the creation of multiple competing consolidators.[616]

Several of these commenters stated that the proposal lacked support to assume that multiple competing consolidators would enter the market.[617] One commenter stated that the proposal assumes there would be a competitive market but lacks support for its assumption that

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615 See Nasdaq Letter IV at 23–26; NYSE Letter II at 9, 13–18; IDS Letter I at 3, 4, 7–8, 9; STANY Letter II at 6; Data Boiler Letter I at 46; Angel Letter at 18, 20.

616 See IDS Letter I at 3, 7; NYSE Letter II at 3, 13; Nasdaq Letter IV at 25; Equity Markets Association Letter at 3 (quoting NYSE Letter II at 3).

617 See NYSE Letter II at 9, 13–18; IDS Letter I at 3–4. One commenter said that the Commission should have considered the European Union's efforts to create a consolidated tape with competing consolidators, noting that no such competing consolidators have registered. Angel Letter at 20. This commenter said that the proposal's lack of discussion of other jurisdictions as alternatives was a potential violation of the APA. See id. at 21. The Commission does not believe the European Union's experience with developing a consolidated tape is relevant for purposes of this proposal. The market and regulatory structure of the European Union are different than they are in the United States. In December 2019, the European Securities and Markets Authority ("ESMA") released a report describing the obstacles to developing a consolidated tape in the EU, including the lack of data quality for OTC transactions, the need for a consolidated tape provider to have to negotiate contracts for data from 170 trading venues and approved publication arrangements, and certain regulatory requirements. See ESMA, MiFID II/MiFIR Review Report No. 1: On the development in prices for pre- and post-trade data and on the consolidated tape for equity instruments (Dec. 5, 2019), available at https://www.esma.europa.eu/sites/default/files/library/mifid_ii_mifir_review_report_no_1_on_prices_for_market_data_and_the_equity_ct.pdf. See also European Commission, The Study on the Creation of an EU Consolidated Tape (Sept. 2020), available at http://www.marketstructure.co.uk/wp-content/uploads/Full-Report--The-Study-on-the-Creation-of-an-EU-Consolidated-Tape.pdf. The U.S. equity markets do not face the same issues and have vast experience in creating consolidated market data.

there would be multiple competing consolidators.[618] This commenter said that a competitive market could not arise if only a few competing consolidators were established, resulting in competing consolidators charging a premium for consolidated market data.[619] The commenter said that the proposal did not consider this possibility, nor did it reasonably consider whether any competing consolidators would register, their viability, and the costs to investors and other market participants if the competing consolidators ceased operating.[620]

One commenter said that there may be few competing consolidators because "SROs or other firms may have cost or other economic advantages (e.g., scale or scope economies) not enjoyed by other potential consolidators…" resulting in competition insufficient to achieve the proposal's goals.[621] Two other commenters, however, stated that the proposal incorrectly

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618 See IDS Letter I at 7. See also NYSE Letter II at 13. This commenter said that the success of the proposed decentralized consolidation model "rests entirely on unfounded assumptions regarding the appearance of a market for competing consolidators…" Id. at 3.

619 See IDS Letter I at 7. This commenter, a market data aggregation firm, also said that it would be very costly for it to become a competing consolidator because it would have to develop a new infrastructure to collect, consolidate, and disseminate NMS data since its method of data consolidation and dissemination is "fundamentally different" than that used by the exclusive SIPs. See IDS Letter II at 1, 2–3.

620 See IDS Letter I at 3, 9. The commenter also said that the Commission failed to meet its burden to examine economic costs and inefficiencies of the proposal because it did not consider the possibility of a delayed implementation, or that it may never be implemented or that it may cease to be viable. See also IDS Letter II at 3. The commenter stated that the proposal did not discuss contingencies in the event of such occurrences. See IDS Letter I at 8. See also Nasdaq Letter IV at 24 (stating that the Commission did not imagine an environment in which only a few competing consolidators survive the initial period of entry).

621 Nasdaq Letter IV at 25. See also Angel Letter at 20 (stating the only competing consolidators will be the two existing exclusive SIPs because only they can afford to comply with Regulation SCI).